Meeting 1210

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December 10, 2025

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12.03 Discussion

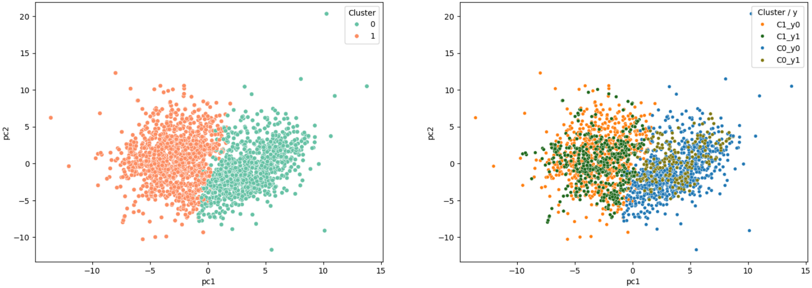
Semiconductor Manufacturing Process

- Given the fitted model $y = \hat{f}$
  - Describe what *hatf* is when x belongs to neighbors of X_0 for some "interesting" X_0
- 研究 $\hat{f}(x), \quad x \in \mathcal{N}(X_0)$ 以理解模型在局部的解釋性、結構。
  - 建構 $\hat{f}$ 的方法（回歸或機器學習），使用鄰近資料 (y_k, x_k) 建立局部模型

12.03 Todo

- Do the SMOTE before PCA after delete outliers
- Decide a x_k(p dimension) from neighbors of X_0 to do the regression
  - 在每個 cluster 抽幾個代表點 X ，例如：
    1. cluster center（在 PCA 空間的重心，找最近的樣本）
    2. y=1 且預測機率高的點
    3. 預測錯誤的點（不穩定區）

Results I SMOTE + PCA + KMeans



cluster	0	1
$y=0$	743	687
$y=1$	205	510

Results I 5-folds cv - mean

model	ACC	FAR	ROCAU
LR	0.8520	0.1011	0.5459
SVM	0.9304	0.0041	0.5027
DT	0.8149	0.1415	0.5309
RF	0.9304	0.0048	0.5074
XGB	0.9272	0.0096	0.5145

model	C_0ACC	C_0FAR	C_0ROCAU	C_1ACC	C_1FAR	C_1ROCAU
LR	$*0.9219$	$*0.0942$	$*0.9431$	$*0.9131$	0.1456	$*0.9233$
SVM	$*0.9895$	$*0.0013$	$*0.9774$	$*0.9883$	0.0044	$*0.9870$
DT	$*0.9114$	$*0.0686$	$*0.8852$	$*0.8622$	0.1716	$*0.8681$
RF	$*0.9631$	0.0000	$*0.9146$	$*0.9749$	$*0.0014$	$*0.9708$
XGB	$*0.9799$	0.0000	$*0.9537$	$*0.9741$	0.0248	$*0.9739$

The End

Questions & Comments